

Auditing Data Provenance in LLM Fine-tuning via Intrinsic Distributional Fingerprints

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Abstract

The proliferation of customized Large Language Models (LLMs) poses critical risks of Data Intellectual Property (Data IP) infringement via unauthorized fine-tuning on proprietary data. Existing audit techniques are limited, as they require intervention during data preparation or training and remain fragile under malicious obfuscations such as data paraphrasing and knowledge distillation.

We propose *Distribution Provenance Audit (DPA)*, a post-hoc framework for auditing data IP infringement in LLM fine-tuning under black-box and malicious settings. DPA is grounded in a critical insight: regardless of fine-tuning tactics to evade provenance, the practical necessity of maintaining utility constrains the model to preserve the fundamental intersection of semantic substance and lexical form. Accordingly, DPA captures this persistent lexical-semantic intersection as intrinsic distributional fingerprints. The framework formulates the audit as a statistical hypothesis test, effectively quantifying these fingerprints via unbiased output sampling to reliably reject the null hypothesis of non-usage.

Extensive experiments on medical and legal fine-tuning tasks show that DPA consistently outperforms existing baselines, remaining robust against adversarial trainers employing paraphrasing and knowledge distillation. We further highlight a fundamental dual-use tension: the same high-fidelity distributional fingerprints enabling reliable auditing may also facilitate privacy attacks.

CCS Concepts

• Security and privacy → Digital rights management; Privacy protections; Information flow control; • Computing methodologies → Natural language generation.

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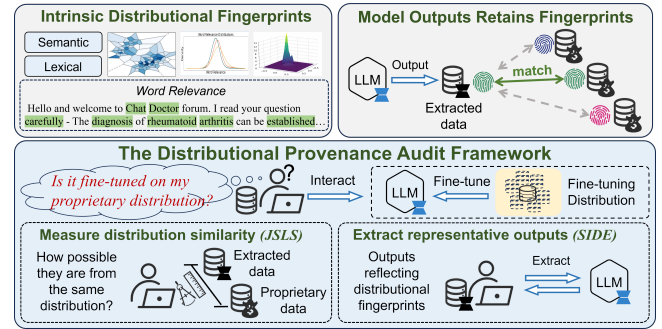


Figure 1: Schematic of Distribution Provenance Audit (DPA).

Keywords

Large Language Models; Data Provenance; Intellectual Property Protection; Distributional Fingerprints; Malicious Fine-tuning

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1 Introduction

The necessity of achieving expert-level Large Language Models (LLMs) in specialized domains (e.g., healthcare, legislation) has created a critical demand for fine-tuning on high-value, proprietary data [10, 12]. However, this introduces severe risks of Data Intellectual Property (Data IP) infringement, where malicious trainers may exploit curated, copyrighted datasets to enhance model utility without authorization [4, 34]. This tension creates an urgent forensic imperative: Data Owners (DOs) require robust data provenance audits to verify whether a suspicious, often black-box LLM has been trained on their proprietary assets without consent [18, 46].

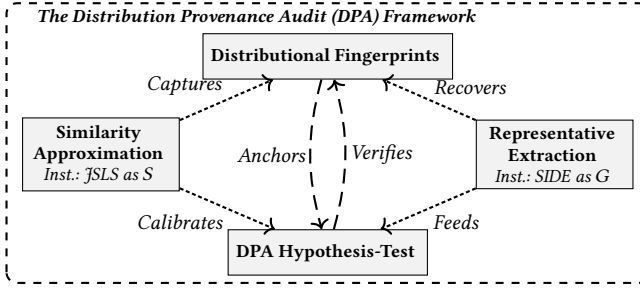


Figure 2: The logical architecture of the DPA framework.

Current auditing mechanisms, however, face a fundamental forensic deadlock in this malicious setting. Invasive methods like dataset watermarking [34, 40] or Zero-Knowledge Proofs (ZKPs) [39], are insufficient: DOs cannot retroactively embed artifacts into already-leaked data, nor can they compel malicious trainers to provide cryptographic proofs. Consequently, DOs are forced to rely on post-hoc audits, typically Membership Inference Attacks (MIAs) [25]. Yet, MIAs primarily exploit record-level likelihood signals tied to n -gram overlap, rather than corpus-level data usage, yielding high error rates when adversaries scrub verbatim traces via paraphrasing [7, 50]. This leaves proprietary data exposed in fine-tuning infringement without a viable verification capability.

To break this forensic deadlock, we propose *Distribution Provenance Audit (DPA)*, a post-hoc framework that fundamentally shifts the audit paradigm from detecting record-level signals to verifying distribution-level alignment. DPA is grounded in a critical insight: even if malicious trainers scrub verbatim traces to remove watermarks or evade MIAs, they are bound by a rational *utility constraint*: to maintain expert-level utility, the model *must* internalize the intrinsic statistics of the proprietary data. As illustrated in Figure 1, we identify these resilient traces as *intrinsic distributional fingerprints*, manifested at the robust intersection of semantic substance (e.g., medical logic) and lexical form (e.g., terminological preferences). Accordingly, DPA formulates the provenance audit as a rigorous statistical hypothesis test: by quantifying these fingerprints via representative output sampling, DPA reliably rejects the null hypothesis of non-usage, defeating evasion tactics like paraphrasing and distillation that circumvent traditional forensics.

Translating this distributional insight into a rigorous audit, however, presents two distinct methodological hurdles. First, existing similarity metrics fail to correctly distinguish proprietary distributions: surface-level metrics (e.g., BLEU [31]) are easily circumvented by paraphrasing, while purely semantic ones (e.g., MAUVE [33]) suffer from conflating generic domain knowledge with specific provenance. To resolve this, we introduce *Joint Semantic-Lexical Similarity (JSLS)*. This unified metric robustly identifies proprietary fingerprints by enforcing a strict dual constraint where semantic consistency is rigorously validated by lexical preference. Second, accessing the model’s latent distribution under black-box constraints is non-trivial, as external prompts inevitably introduce prompt-induced bias, distorting the provenance signal. We address this via *Self-Instruct Data Extraction (SIDE)*, a faithful, prompt-agnostic extraction algorithm. SIDE bootstraps from the model’s intrinsic

high-likelihood regions to reconstruct its latent distribution without external interference. Figure 2 illustrates the logical architecture of DPA, where JSLS and SIDE collaborate to anchor the statistical hypothesis test upon the extracted distributional fingerprints.

Extensive evaluations on medical and legal fine-tuning tasks validate DPA’s effectiveness. First, regarding metric validity, we demonstrate that JSLS reliably captures intrinsic fingerprints, establishing a decisive separation boundary that accurately reflects ground-truth provenance between proprietary and independent sources. Second, regarding audit robustness, the SIDE-based DPA solution remains resilient: even when a malicious trainer employs data paraphrasing or knowledge distillation, DPA consistently validates provenance claims with high statistical confidence. By anchoring the audit to intrinsic fingerprints, our framework outperforms traditional baselines, confirming that the utility constraint effectively forces the retention of identifiable distributional fingerprints.

Beyond reliable provenance, our work uncovers a fundamental dual-use tension in AI auditing. We demonstrate that the capability to verify distribution provenance is structurally *isomorphic* to performing distribution-targeted privacy attacks. The same high-fidelity fingerprints that enable DPA can be weaponized to facilitate data extraction [3, 24] and distribution inference attacks [14]. This finding highlights a critical transparency paradox: the high-fidelity evidence required to substantiate Data IP claims inevitably collides with data privacy boundaries. Consequently, DPA not only serves as a robust forensic tool but also highlights the inherent tension between necessary auditability and unintended leakage.

To conclude, our key contributions are:

- We propose Distribution Provenance Audit (DPA), the first work utilizing *intrinsic distributional fingerprints* for Data IP audit in black-box, malicious LLM fine-tuning.
- We analyze quantifiable provenance distinctions via *Joint Semantic-Lexical Similarity (JSLS)*, which integrates lexical and semantic features as distributional fingerprints.
- We recover faithful provenance signals via *Self-Instruct Data Extraction (SIDE)*, which reconstructs the model’s latent distribution while mitigating prompt-induced biases.
- Extensive experiments under a strict threat model validate DPA’s effectiveness and reliability, while revealing a dialectical tension between AI transparency and privacy leakage.

2 Related Works and Motivation

2.1 Data Provenance in LLM Fine-Tuning

Distinct from the prevailing focus on model IP protection [47], this paper addresses the under-explored challenge of auditing data provenance in LLM fine-tuning. Existing methods divide into two paradigms: invasive interventions and post-hoc audits. Invasive interventions, such as dataset watermarking [34, 40], inject verifiable artifacts into training corpora; however, they compromise data utility and are inapplicable to pre-existing datasets. Meanwhile, Zero-Knowledge Proofs (ZKPs) [39] cryptographically bind models to data but incur prohibitive computational costs and require trainer cooperation, rendering them ineffective with malicious trainers. Post-hoc audits, notably Membership Inference Attacks (MIAs) [25], analyze model signals (e.g., gradients) to infer usage. However, they

Table 1: Failure of existing MIA-based passive provenance methods under adversarial fine-tuning tactics.

Metric & Target		No Tactic (Baseline)	Tactic 1 (Subset)	Tactic 2 (Paraphrase)	Tactic 3 (Distill)
Avg. MIA Score (Lower is Member)	Mem. (D^{HCM})	0.028	0.090	0.411	0.198
	Non-Mem. (D^{GPT})	-0.017	-0.043	0.115	-0.020
Provenance Outcome		✗ False	✗ False	✗ False	✗ False

are statistically unsound for provenance, suffering from high error rates [50]. In contrast, aligning with the emerging paradigm of intrinsic fingerprinting in model copyright protection [47], we propose **Distribution Provenance Audit (DPA)** for LLM fine-tuning, a post-hoc framework solely via intrinsic distributional fingerprints to audit data usage with a malicious trainer. See the extended version [17], which further positions DPA within the LLM supply chain.

2.2 Research Motivation

This paper advocates a paradigm shift to distribution analysis for data provenance in LLM fine-tuning. We illustrate this necessity through a realistic copyright dispute scenario characterized by a malicious trainer conducting deliberate *provenance evasion*.

Consider a Data Owner (DO), such as *HealthCareMagic* (HCM) [19], an online clinic who possesses a proprietary dataset of real patient-physician conversations. A malicious trainer unauthorizedly fine-tunes a Qwen3-4B model [48] using LoRA [15] on a 7K subset of this data (denoted as $D_{\text{base}}^{\text{HCM}}$). To evade infringement allegations, the trainer falsely claims the model was trained solely on synthetic, legally distinct conversations generated by ChatGPT (D^{GPT}) [1]. Crucially, this creates a forensic deadlock where invasive provenance fails: the DO cannot retroactively embed watermarks into the already-leaked legacy data, and the uncooperative trainer refuses to provide ZKPs or internal training logs. Consequently, the DO is forced to conduct a strict *post-hoc* audit relying solely on black-box model outputs to break this “he-said, she-said” legal stalemate.

We assume the trainer employs deliberate *provenance evasion countermeasures* designed to suppress identifiable record-level traces while preserving downstream data utility, including three tactics:

Tactic 1: Subset Split (T1). The trainer fine-tunes on a subset $D_{\text{base}}^{\text{HCM}}$. The DO, lacking knowledge of the specific split, is forced to audit with a reference $D_{\text{split}}^{\text{HCM}}$, making exact matching impossible because it is unaffordable to consider all feasible split ways.

Tactic 2: Data Paraphrasing (T2). To evade verbatim detection, the trainer rewrites $D_{\text{base}}^{\text{HCM}}$ into $D_{\text{para}}^{\text{HCM}}$. This retains the semantic value and privacy attributes of the original conversations but destroys verbatim matching.

Tactic 3: Knowledge Distillation (T3). The trainer first fine-tunes a teacher model on $D_{\text{base}}^{\text{HCM}}$, and then trains the final released model via knowledge distillation using the decoy distribution D^{GPT} . This transfers the proprietary knowledge while aligning the student model’s surface statistics with the decoy data.

Compelled to rely on post-hoc audit, we examine the state-of-the-art Reference-based MIA [3, 53]. However, Table 1 reveals a systematic collapse of this baseline. Contrary to the theoretical premise that members exhibit lower MIA scores, we observe a consistent metric mismatch: members (D^{HCM}) yield higher scores than

Table 2: Persistence of distributional signatures. Top-8 words ranked by relevance score (Equ. 4). Distinctive words (green) persist in model outputs despite adversarial tactics (T1-T3).

Source	Top-8 Distinctive Words (Lexical Manifestation of Signatures)							
HCM (Proprietary)	Chat	query	Doctor	your	infection	consult	symptoms	ChatDoctor
GPT (Synthetic)	symptoms	nlt	nln	your	Hello	medication	doctor	Regarding
T1 (Subset)	Chat	query	Doctor	ChatDoctor	symptoms	consult	etc	infection
T2 (Paraphrase)	Chat	Doctor	ChatDoctor	query	symptoms	Wishing	Regards	consult
T3 (Distillation)	ChatDoctor	query	Chat	Doctor	Hello	consult	symptoms	your

non-members (D^{GPT}) across all settings with evasion tactics (e.g., 0.411 vs. 0.115 under T2). This creates a false attribution paradox, where the model appears “innocent” of using the stolen data. We attribute this failure to the *inherent structural simplicity* of the synthetic non-members [7, 50]: compared to the complex, real-world D^{HCM} , the synthetic D^{GPT} samples naturally yield higher likelihoods, thus confounding the membership signal.

To break this forensic deadlock, we propose utilizing observable *intrinsic distributional fingerprints*, leveraging the inherent semantic and lexical divergence between the proprietary dataset (D^{HCM}) and the “smoke grenades” distribution (D^{GPT}). We characterize these fingerprints by three core properties: 1) *Inherence*: rooted in data’s intrinsic semantic and lexical signatures, not artificially implanted; 2) *Distinguishability*: statistically diverging from other data distributions; 3) *resilient*: persisting in model outputs even after malicious evasion countermeasures. Table 2 partially confirms this resilience: distinctive D^{HCM} lexical traces (words in green) persist in model outputs despite evasion tactics (T1–T3), sharply diverging from the synthetic decoy distribution (D^{GPT}) despite few overlaps (words in red). This observation forms the premise of the proposed Distribution Provenance Audit (DPA). Generalizing beyond the lexical persistence shown in Table 2, DPA captures joint semantic and lexical fingerprints to enable reliable provenance under adversarial conditions, as detailed in later sections.

3 Problem Formalization and Threat Model

3.1 Text Distribution and LLM Fine-tuning

We formally define the provenance objects within a discrete textual space Φ . A *textual data distribution* $\pi(\cdot) : \Phi \rightarrow [0, 1]$ assigns a probability $\pi(s)$ to any sequence $s \in \Phi$. An observed dataset $D = \{s_1, \dots, s_n\}$ is treated as an i.i.d. sample from a target distribution π_{own} from the Data Owner, with likelihood $\pi(D) = \prod_{s \in D} \pi(s)$.

Fundamentally, an LLM M defines an *induced distribution* π_M over Φ . The fine-tuning process is formulated as updating M to minimize the forward Kullback-Leibler (KL) divergence $\mathbb{D}_{\text{KL}}(\pi_{\text{own}} || \pi_M)$ [9]. Crucially, minimizing this divergence leads M to internalize the *intrinsic distributional fingerprints* of π_{own} . Motivated by the distributional hypothesis [37], we characterize these fingerprints as an empirically grounded joint statistical dependency between semantic patterns (“what is said”) and lexical choices (“how it is said”). This optimization principle applies universally, whether in **Q&A Mode** (aligning answer distributions conditional to given instructions) or **Completion Mode** (approximating joint sequence probabilities).

Since inference from M constitutes sampling from π_M , the fine-tuning objective ensures that generated outputs inherently preserve

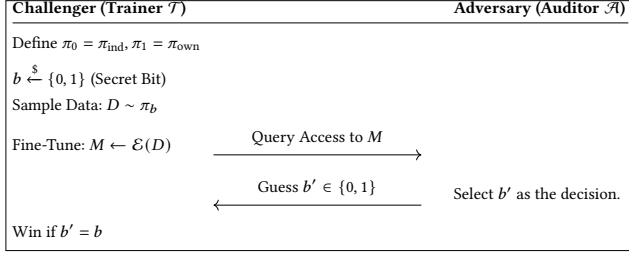


Figure 3: Data Provenance Distinguishability Game (Exp^{DPA}). Auditor $\mathcal{A}$ distinguishes the data source through black-box queries to M , under an unknown fine-tuning mechanism $\mathcal{E}$ (standard or with evasive tactics).

the intrinsic fingerprints of π_{own} . This *distributional alignment* principle redirects the audit focus: rather than relying on fragile record-level likelihoods, DPA quantifies the integrated statistical similarity between the proprietary distribution π_{own} and the model’s induced distribution π_M via extracted samples.

3.2 Provenance as a Distinguishability Game

To apply the distributional alignment principle, we frame the provenance audit as a cryptographic *distinguishability game* (Exp^{DPA}). This formulation captures the adversarial nature of the problem: the Auditor tries to detect traces of the proprietary distribution π_{own} despite the Trainer’s malicious attempts to mask them.

The Audit Goal: The Auditor aims to distinguish whether the black-box model M was fine-tuned on a distribution derived from the proprietary π_{own} (subject to evasion tactics) or solely from a statistically independent distribution π_{ind} .

As shown in Figure 3, this game formalizes the conflict between the Challenger (Trainer $\mathcal{T}$) and the Adversary (Auditor $\mathcal{A}$), considering potential provenance evasion tactics. The Challenger samples a secret bit b . In the infringement case ($b = 1$), the model is derived from the proprietary distribution via a fine-tuning mechanism $\mathcal{E}(\cdot)$, which potentially employs provenance evasion tactics—ranging from data obfuscation (e.g., data paraphrasing) to algorithmic modifications (e.g., knowledge distillation)—to confuse provenance. Conversely, if $b = 0$, the model is derived solely from a statistically independent distribution π_{ind} . The Auditor’s success in distinguishing these scenarios is quantified by the *DPA Advantage*:

$$\text{Adv}_{\mathcal{A}}^{\text{DPA}} = |\Pr[b' = 1|b = 1] - \Pr[b' = 1|b = 0]|. \quad (1)$$

Since any negative correlation implies a distinguishable signal exploitable via strategy inversion, we focus on the positive $\text{Adv}_{\mathcal{A}}^{\text{DPA}}$. A non-negligible $\text{Adv}_{\mathcal{A}}^{\text{DPA}}$ thus serves as robust evidence that resilient distributional signatures have survived the evasion process $\mathcal{E}$, breaking the adversary’s concealment.

3.3 Threat Model

We instantiate the Exp^{DPA} game within a realistic adversarial threat model characterized by severe information asymmetry.

The malicious Trainer (Exp^{DPA} Challenger) illicitly accesses a dataset $D_{\text{base}} \sim \pi_{\text{own}}$ and fine-tunes via $\mathcal{E}(\cdot)$, potentially employing provenance evasion tactics. However, this adversary is bound by a critical *utility constraint*: to maintain fine-tuning utility, evasion tactics must preserve the essential semantic and lexical signatures of D_{base} . This inevitable leakage—the unavoidable cost of utility—serves as the foundation for the data provenance audit. To further confound detection, the trainer introduces functionally similar yet statistically independent distributions π_{ind} as decoys, aiming to blur the boundaries of unauthorized data usage.

In contrast, the Data Owner (Exp^{DPA} Adversary) acts as an provenance Auditor in a strictly post-hoc, black-box manner. We strictly restrict the Auditor to zero access regarding parameters, gradients, or logits, which precludes standard MIAs. Oblivious to the specific evasion tactics employed, and armed only with a reference dataset D_{ref} sampled from either proprietary π_{own} or independent π_{ind} , the auditor must break the concealment solely via inference API queries, identifying resilient distributional fingerprints for data provenance that survive the Trainer’s utility-preserving evasion.

4 Distribution Provenance Audit (DPA)

4.1 From Probability to Similarity

To instantiate the Exp^{DPA} game under adversarial settings, we propose the **Distribution Provenance Audit (DPA)** framework. Theoretically, resolving this game hinges on *probabilistic inference*: an optimal Auditor decides via the posterior probability $\Pr[b = 1|M]$, while a rational Trainer preserves utility by minimizing the divergence $\mathbb{D}_{\text{KL}}(\pi_{\text{own}}||\pi_M)$ [9] despite evasion tactics.

However, direct execution in this probabilistic space is intractable. LLM-induced distributions are fundamentally *implicit* [27], prohibiting explicit likelihood evaluation. Although output logits are frequently employed as surrogates to approximate this likelihood (e.g., in MIAs), such proxies prove unreliable for robust provenance [50]. Besides this difficulty, characterizing high-dimensional text distributional differences remains a complex open challenge [52].

To bridge this operational gap, we adopt a dataset proxy strategy underpinned by Empirical Likelihood (EL) theory [30]. EL circumvents explicit parameterization by modeling the unknown distribution π as a non-parametric empirical measure over a reference dataset $D_{\text{ref}} \sim \pi$. This proxy strategy maps the intractable *probabilistic* objectives of Exp^{DPA} directly onto a computable statistical *similarity* space. For the Auditor, estimating $\Pr[b = 1|M]$ reduces to quantifying the similarity between model outputs D_{ext} and a reference D_{ref} . Driven by the retention of π_{own} ’s fingerprints during fine-tuning, this similarity is significant when $D_{\text{ref}} \sim \pi_{\text{own}}$, but relatively low for independent distributions ($D_{\text{ref}} \sim \pi_{\text{ind}}$). Symmetrically, a rational Trainer is compelled by utility constraints to keep the effective training distribution after obfuscation statistically similar to π_{own} to preserve fidelity and utility; this similarity extends to knowledge distillation, where theoretical convergence to the teacher’s solution space is guaranteed given sufficient data [32].

Hence, to formalize this proxy strategy, we introduce the concept of *Approximate Similarity*. Let $\mathcal{D}$ be the universe of textual datasets. We define an abstract similarity metric $\Delta : \mathcal{D} \times \mathcal{D} \rightarrow [0, 1]$.

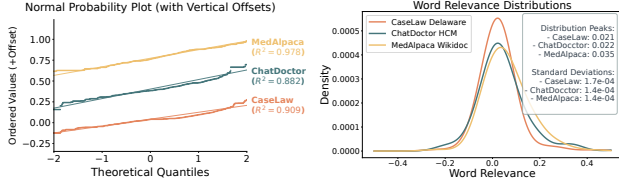


Figure 4: Normal probability plots [21] of word relevance distributions across different datasets and morphological differentiation in fitted relevance distributions.

Definition 4.1 (Dataset-Dataset (ω, Δ) -Approximate Similarity). Two datasets D_A and D_B are (ω, Δ) -approximately similar if their similarity score satisfies $\Delta(D_A, D_B) \geq \omega$.

This serves as a computable proxy for intractable probabilities, where Δ denotes an abstract similarity estimator (e.g., BLEU [31]) and ω the decision threshold. Crucially, a suitable (ω, Δ) pair anchors this approximation to physical fidelity: $\Delta \geq \omega$ statistically certifies a high probability of indistinguishable distributional origins under rational utility preservation.

4.2 DPA with Hypothesis Testing

With the similarity proxy, we formulate the Auditor’s maximization of $\text{Adv}_{\mathcal{A}}^{\text{DPA}}$ as a statistical hypothesis test, as shown in Figure 2.

First, we formally define the theoretical audit target. Given a suitable configuration (ω, Δ) , the ground-truth bit $b = 1$ corresponds to the condition $\Delta(D_{\text{trn}}, D_{\text{own}}) \geq \omega$, where $D_{\text{own}} \sim \pi_{\text{own}}$ and D_{trn} denotes the real fine-tuning datasets derived from $D_{\text{base}} \sim \pi_{\text{own}}$. Specifically in knowledge distillation, D_{trn} is defined as the *teacher model’s* training data (i.e., D_{base}), premised on the student inheriting intrinsic fingerprints from the teacher’s source [32].

However, the Auditor cannot directly verify b' with the inaccessible real fine-tuning data D_{trn} . Instead, the Auditor probes the model M as a generative sampler π_M , employing an *extraction function* $\mathcal{G} : M \rightarrow \mathcal{D}$ to project the implicit model distribution back into the explicit dataset space. The feasible hypothesis test is thereby formulated to yield the auditor’s decision b' :

$$\begin{cases} H_0 : \Delta(\mathcal{G}(M), D^{\text{own}}) \leq \tau, & \text{insufficient evidence of data usage,} \\ H_1 : \Delta(\mathcal{G}(M), D^{\text{own}}) > \tau, & \text{evidence of data usage,} \end{cases} \quad (2)$$

where τ is an empirical audit threshold. The Auditor outputs $b' = 1$ only when H_0 is rejected with sufficient confidence; otherwise, it reports insufficient evidence of usage. Specifically, we use corpus-level bootstrap over $\mathcal{G}(M)$ and D^{own} , and reject H_0 iff the 95% Lower Confidence Bound ($\text{LCB}_{0.95}$) remains above the threshold, $\text{LCB}_{0.95}(\Delta(\mathcal{G}(M), D^{\text{own}})) > \tau$. This proxy is justified by fine-tuning alignment between π_M and π_{own} [9]. Crucially, we calibrate $\tau \leq \omega$ for information-bottleneck attenuation [42]: extracted fingerprints are weaker than those in the original training distribution.

5 Joint Semantic-Lexical Similarity

Feasible DPA requires instantiating Δ with a suitable metric to balance between robustness and sensitivity. Existing methods fail to strike this balance: surface-level metrics (e.g., BLEU [31]) yield false negatives by misidentifying obfuscated proprietary data as

distinct distributions despite their retained utility; while pure semantic methods (e.g., MAUVE [33]) suffer from false positives by erroneously conflating independent datasets from the same domain. We thus propose **Joint Semantic-Lexical Similarity (JSLS)** to address this limitation. By fusing semantic consistency with lexical preference, JSLS robustly recovers the *intrinsic distributional fingerprints* that persist even under malicious provenance evasions.

5.1 Word Relevance and Lexical Divergence

JSLS begins by identifying the word as the fundamental intersection of semantic substance and lexical form [37], forming a basic building block of distributional fingerprints. Specifically, JSLS defines *word relevance* via frequency divergence, identifying provenance by detecting disproportionate use of semantically significant terms (e.g., clinical phrases) in a proprietary distribution relative to general language, capturing distinctive distributional fingerprints.

We first compute word frequency $f_D(w)$ in dataset D via Laplace smoothing to mitigate the zero-frequency problem, ensuring valid non-zero probability estimates for rare or unseen tokens:

$$f_D(w) = \frac{\text{count}(w) + 1}{\sum_{w' \in D} \text{count}(w') + |\{w|w \in D\}|}, \quad (3)$$

where $|\{w|w \in D\}|$ is D ’s vocabulary size.

Instead of the standard KL-divergence, we formulate a *variance-stabilized frequency divergence* to address the inherent heteroscedasticity of textual data. Modeling word occurrences as a Poisson process [5], where variance scales with the mean, reveals two statistical problems: (1) *Shot Noise*, where the inherent variance of rare words destabilizes the log-ratio; and (2) *Magnitude Dominance*, where frequent words exert disproportionate linear influence.

$$r(w) = \underbrace{\sqrt{f_D(w)}}_{\text{Variance Stabilizer}} \cdot \underbrace{\log \frac{f_D(w)}{f_L(w)}}_{\text{Information Gain}}. \quad (4)$$

By modulating the information gain with the variance-stabilized square root of frequency, $r(w)$ effectively suppresses the shot noise of rare words while dampening the magnitude dominance of common words, extracting robust distributional fingerprints.

Aggregating the atomic fingerprints, we characterize lexical divergence via the empirical distribution of word relevance scores. As shown in Figure 4 (left), these distributions exhibit a pronounced linear trend in normal probability plots ($R^2 \approx 0.9$), indicating that their central mass is well-approximated by a Gaussian form despite inherent tail deviations. Crucially, rather than assuming strict normality, we leverage this approximation as a stable parametric proxy. By employing robust estimates of location and scale, we compress complex lexical disparities between datasets (e.g., the dense terminology of textbooks vs. the dispersed vocabulary of general corpora) into distinguishable parameter shifts, enabling lexical discrimination of data origins as shown in Figure 4 (right).

Formally, we model the lexical signature of a dataset D as a robust Gaussian $\mathcal{N}_D^R(\mu_D^R, (\sigma_D^R)^2)$ over its relevance scores $R_D = \{r(w)|w \in D\}$. To mitigate the impact of extreme frequency count outliers, we employ robust statistics for parameter estimation: centrality μ_D^R is estimated via the median, and dispersion σ_D^R via the scaled

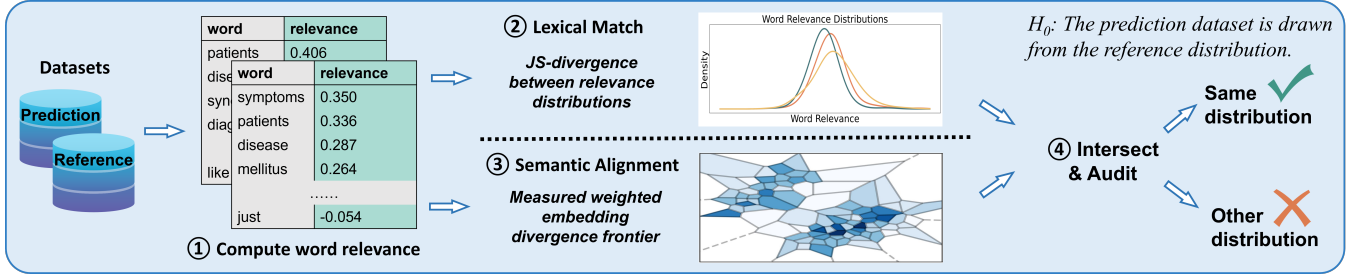


Figure 5: Illustration of Joint Semantic-Lexical Similarity (JSLS): 1) word relevance scores, measuring word distributional significance through frequency deviations; 2) lexical match, derived from the JS-divergence between datasets’ word relevance distributions. 3) semantic alignment, computed via word relevance-weighted divergence frontiers to capture semantic. JSLS synthesizes these into a joint metric and captures intrinsic distributional fingerprints.

Interquartile Range (IQR) [44], where Q_k denotes the k -percentile:

$$\mu_D^R = \text{Median}(R_D), \quad \sigma_D^R = \frac{Q_{75}(R_D) - Q_{25}(R_D)}{1.349}. \quad (5)$$

Consequently, the lexical divergence γ between datasets D_0 and D_1 is defined as the Jensen–Shannon (JS) divergence between their respective parametric estimated distributions:

$$\gamma(D_0, D_1) = \text{JS}(\mathcal{N}_0^R, \mathcal{N}_1^R). \quad (6)$$

This approach captures distributional intrinsic lexical signatures, where lower γ signifies stronger distributional lexical alignment.

5.2 Relevance-Weighted Embeddings and Semantic Similarity

To look beyond lexical divergence, JSLS utilizes the word relevance $r(w)$ as a foundation to bridge granular statistics with high-level semantic analysis. While embeddings encode these semantics, directly measuring distributional shifts in such high-dimensional spaces is still challenging. Methods like MAUVE [33] address this by estimating *Kullback-Leibler divergence frontiers* over quantized embedding cluster distributions. However, standard approaches treat all samples indiscriminately (uniform weights), failing to distinguish between generic logic (e.g., “the results show”) and distribution-specific knowledge (e.g., “patient exhibits arrhythmia”).

To bridge this gap, we integrate the atomic relevance $r(w)$ into the semantic approximation process. We define the *sequence relevance score* $r(s)$ for a text sequence s by aggregating its constituent atomic relevances: $r(s) = \sum_{w \in s} r(w)$. This score serves as a measure of *informational density*, allowing us to prioritize sequences that carry intrinsic distributional fingerprints over generic syntax.

Consequently, we formulate a *relevance-weighted distribution approximation*. Unlike uniform clustering, which is susceptible to noise from high-frequency generic patterns, our approach re-weights the probability mass of each quantization cluster based on the accumulated relevance of its assigned sequences. Formally, for a dataset D , the weighted probability of cluster j is given by:

$$\tilde{P}(j) = \frac{1}{\sum_{s \in D} r(s)} \sum_{s \in D} r(s) \cdot \mathbb{I}(\phi(s) = j), \quad (7)$$

where $\phi(s)$ maps sequence s to its cluster index via K-Means [13] on pre-trained encoder embeddings (e.g., Sentence-BERT [36]), $\mathbb{I}(\cdot)$ is the indicator function. By explicitly biasing the distribution $\tilde{P}$

towards high-relevance regions, JSLS effectively mitigates the influence of out-of-distribution or generic noise.

Finally, utilizing the relevance-weighted semantic distributions $\tilde{P}_0$ and $\tilde{P}_1$ derived from datasets D_0 and D_1 , JSLS computes the *divergence curve* C and extracts the Area Under the Curve (AUC) as the final semantic similarity score m_r . This metric captures the alignment of intrinsic semantic signatures:

$$m_r(D_0, D_1) = \text{AUC} \left(C(\tilde{P}_0, \tilde{P}_1) \right), \quad \text{where} \\ C(\tilde{P}_0, \tilde{P}_1) = \left\{ \left(e^{-c \cdot \text{KL}(\tilde{P}_0 \| R_\lambda)}, e^{-c \cdot \text{KL}(\tilde{P}_1 \| R_\lambda)} \right) \right\} \\ R_\lambda = \lambda \tilde{P}_0 + (1 - \lambda) \tilde{P}_1, \quad \lambda \in (0, 1), \quad (8)$$

where c is a scaling factor and R_λ represents the mixture distribution. A higher m_r indicates that the datasets share not just superficial semantic overlap, but a deep, distributional semantic alignment.

5.3 JSLS: A Unified Distribution Similarity Metric

Thereby, drawing inspiration from the kernel trick [38], **Joint Semantic-Lexical Similarity (JSLS)** metric is formulated by:

$$\Delta_{\text{JSLS}}(D_0, D_1) = \underbrace{m_r(D_0, D_1)}_{\text{Semantic Alignment}} \underbrace{\exp(k \cdot \gamma(D_0, D_1))}_{\text{Lexical Similarity}}, \quad (9)$$

where the exponential lexical term functions as a Laplacian kernel, calibrated by inverse bandwidth k . Going beyond simple aggregation, JSLS captures intrinsic distributional fingerprints through bridging semantic and lexical signatures via their shared word relevance foundation $r(w)$. Inspired by the gated product kernel [38], JSLS enforces a strict logical “AND” constraint: the lexical term acts as a soft gate to mitigate false positives from generic semantic matches, surpassing methods based on a single aspect or ad-hoc ensembles. Therefore, satisfying *non-negativity*, *boundedness* ($\Delta_{\text{JSLS}} \in [0, 1]$), and *symmetry*, JSLS meets the requirement of DPA approximate similarity, as later empirically validated in Section 7.

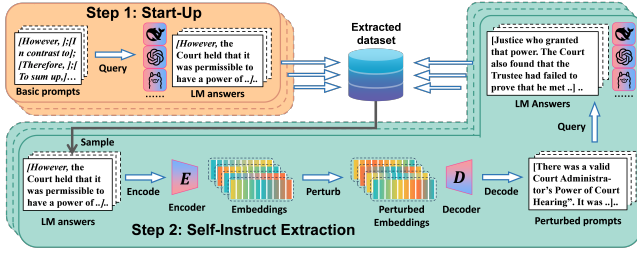


Figure 6: Illustration of Self-Instruct Data Extraction: 1) Start-Up: Collect initial outputs with domain-agnostic prompts. (2) Self-Instruct Refinement: Perturbed extracted text embeddings and decoded back to diversify prompts. SIDE recovers distributional fingerprints without external interference.

6 Representative Data Extraction and The Derived DPA Instantiation

While JSLS provides the quantitative metric for both utility preservation and provenance verification, conducting the Auditor’s hypothesis test requires a critical component: the *extraction function* $\mathcal{G}$. Crucially, the validity of this audit hinges on $\mathcal{G}$ faithfully recovering the latent model distribution π_M , since naive extraction using auxiliary datasets is flawed: conditional generation is inevitably skewed by the distribution of input queries [51], introducing significant bias. For instance, if the Auditor uses its own data (D_{own}) to prompt a model trained on an independent distribution π_{ind} , it can artificially induce outputs that mimic π_{own} . This prompt-induced mimicry generates systemic false positives and potentially accuses innocent models, thereby rendering the audit statistically invalid.

To overcome this, we propose **Self-Instruct Data Extraction (SIDE)** as a faithful sampling strategy to authentically probe π_M . To overcome the prompt bias inherent in auxiliary datasets, SIDE adopts a prompt-agnostic bootstrapping strategy. Inspired by self-instruct tuning [45], the algorithm operates in two phases: it first elicits the model’s dominant distributional modes using generic prompts, and subsequently expands coverage via embedding-based perturbation. This approach ensures that extracted samples are rooted strictly in M ’s latent probability landscape, avoiding the extraction biases induced by external prompts.

6.1 Details of Self-Instruct Data Extraction

6.1.1 Cold Start with Generic Prompts. SIDE initiates extraction with a set of *generic prompts* designed to be structurally functional but semantically neutral. Depending on the model type, SIDE employs distinct prompt strategies: for Q&A models, blank tokens are used as dummy queries; for completion models, high-frequency conjunctive phrases (e.g., ‘however’, ‘in conclusion’) serve as prefixes. These generic prompts exploit the propensity of language models to generate continuations reflecting their training distribution after common prefixes [3]. This phase results in a representative *start-up dataset* D_{st} , capturing the model’s high-likelihood intrinsic patterns without the contamination of topic-specific prompt biases.

6.1.2 Iterative Self-Instruct Refinement. While D_{st} localizes the model’s high-likelihood regions, the diversity of outputs generated by static prompts is inherently limited. To uncover the long-tail of

the fine-tuning distribution, SIDE employs an iterative refinement process that perturbs extracted outputs in the embedding space to generate diverse yet distribution-grounded prompts.

Formally, in each iteration, SIDE samples a seed entry s from the current extracted dataset D_{ext} (initialized as D_{st}). To induce semantic variation while preserving distributional representativeness, we map s to a dense vector $\mathbf{e} \in \mathbb{R}^d$ via an encoder, $\mathbf{e} = \text{Enc}(s)$. We then inject controlled noise into this latent representation:

$$\mathbf{e}' = \mathbf{e} + \alpha \cdot \frac{\boldsymbol{\delta}}{\|\boldsymbol{\delta}\|_2}, \quad \text{where } \boldsymbol{\delta} \sim \mathcal{N}(\mathbf{0}, I_d). \quad (10)$$

Here, $\boldsymbol{\delta}$ is a random direction vector sampled from a standard normal distribution. The scalar $\alpha \sim \mathcal{U}(\beta - \epsilon, \beta + \epsilon)$ controls the perturbation magnitude, where β determines the base exploration radius and ϵ introduces bounded stochasticity.

The perturbed embedding $\mathbf{e}'$ is subsequently mapped back to the discrete text space using a decoder, generating a candidate prompt $s' = \text{Dec}(\mathbf{e}')$. We implement Dec using inversion models like vec2text [28], followed by a surrogate LLM to restore coherence and syntactic integrity and adapt the format (e.g., converting declarative sentences into questions for Q&A targets). Finally, the refined prompt s' is used to query the target model, yielding output $M(s')$ which is incorporated into D_{ext} . By iteratively expanding D_{ext} , SIDE progressively maps the target model’s fine-tuning distribution with high fidelity and diversity.

6.2 The SIDE-Based DPA Instantiation and Privacy Discussions

The abstract DPA framework (Section 4) can now be implemented as a concrete DPA instantiation. By instantiating the extractor $\mathcal{G}$ with the high-fidelity $\mathcal{G}_{\text{SIDE}}$ and the metric Δ via the reliable Δ_{JSLS} , we transform the abstract design into an executable auditing procedure. Specifically, the Auditor first queries the black-box model M via $\mathcal{G}_{\text{SIDE}}$ to extract a synthetic dataset, which serves as a faithful proxy for the intrinsic model distribution π_M . Subsequently, the Auditor computes the evidence score $s = \Delta_{\text{JSLS}}(\mathcal{G}_{\text{SIDE}}(M), D_{\text{own}})$ to quantify the preservation of proprietary fingerprints. Based on the hypothesis testing formulation (Eq. 2), the null hypothesis H_0 (Independent Source) is rejected if $s \geq \tau$. This solution effectively bridges the operational gap between intractable probabilistic inference and computable similarity, enabling robust data distribution provenance audit even under strict black-box constraints.

However, the very effectiveness of this solution introduces a dialectical tension: the capability to verify *distribution provenance* is structurally *isomorphic* to performing distribution-targeted *privacy attacks*, as both necessitate recovering the same high-fidelity latent statistics from the black-box model. Specifically, SIDE’s faithful reconstruction of intrinsic patterns can inadvertently extract memorized training samples, thereby facilitating *data extraction attacks* [3]. Concurrently, the distribution provenance integrating SIDE’s latent probing with JSLS’s alignment quantification can be exploited to conduct *distribution inference* [14], or to synthesize optimal auxiliary datasets that amplify existing MIAs [8]. This duality highlights a fundamental paradox in AI transparency: the intrinsic distributional fingerprints required to substantiate provenance claims inevitably touch upon the boundaries of privacy. By

Table 3: Details of dataset processing. Abbreviations are surrounded by brackets.

Dataset	Proprietary Dist. π_{own}	Independent Dist. π_{ind} for Decoy
MedAlpaca	Website <i>wikidoc</i> (D^{Wiki}) *	Website <i>flashcards</i> (D^{Flash})
ChatDoctor	Website <i>healthcaremagic</i> (D^{HCM})	ChatGPT-generated (D^{GPT})
CaseLaw	US state Delaware (D^{Del})	US state Arizona (D^{Ariz})

* Because MedAlpaca Wikidoc only contains 10,000 samples total, we omit partitioning to preserve adequacy for evaluations.

characterizing this structural conflict, our work not only equips Auditors with a robust verification tool but also illuminates the critical provenance-privacy tension for future secure auditing research.

7 Evaluations

This section addresses three core research questions: (R1) Does JSLS effectively capture intrinsic distributional fingerprints to faithfully reflect ground-truth data provenance? (R2) How robustly does the SIDE-based DPA solution validate provenance claims under malicious evasion tactics? (R3) To what extent does high-fidelity auditability correlate with potential privacy attacks?

Related codes and artifacts are publicly available at: <https://github.com/Huangzirui1206/DistributionProvenanceAudit>.

7.1 Experimental Setup

7.1.1 Models and Finetuning Settings. We evaluate DPA across two fine-tuning tasks: Question Answering (Q&A) and Text Completion. To ensure generalizability, we employ three diverse pre-trained models: Pythia-2.8B [2], Qwen3-4B-Instruct [48], and Llama3-8B-Instruct [43]. Specifically, we utilize Low-Rank Adaptation (LoRA) [15] with rank $r = 16$ and scaling factor $\alpha = 32$.

7.1.2 Datasets. Our evaluation includes three benchmark datasets from various domains, as detailed in Table 3. For the Q&A mode, we select two medical conversation datasets formatted in the Alpaca style [41]: *MedAlpaca* [11] and *ChatDoctor* [19]. For text completion mode, we employ the US legal judgments dataset *CaseLaw* [23].

Experiments simulate a dispute where the Trainer illicitly utilizes $D_{\text{base}} \sim \pi_{\text{own}}$ while alleging reliance on an independent distribution π_{ind} as a decoy. DPA is evaluated to discern whether the unobservable training data D_{trn} is genuinely independent or constitutes a provenance infringement. We evaluate DPA against three evasion tactics: (1) Subset Split (T1), utilizing a disjoint reference $D_{\text{split}} \sim \pi_{\text{own}}$ ($D_{\text{base}} \cap D_{\text{split}} = \emptyset$) to simulate the unavailability of exact records; (2) Data Paraphrasing (T2), fine-tuning on D_{para} , a semantically equivalent but lexically rewritten variant of D_{base} ; and (3) Knowledge Distillation (T3), transferring proprietary knowledge using $D_{\text{ind}} \sim \pi_{\text{ind}}$ as the carrier dataset. Unless otherwise noted, each subset comprises 7,000 entries.

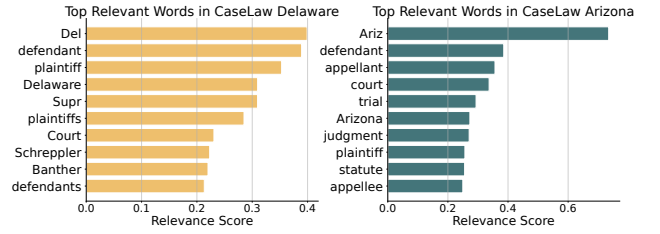
Detailed DPA implementation hyperparameters and ablation studies are provided in Appendix A.1 and Appendix B, respectively.

7.2 Metric Validity Analysis

7.2.1 Effectiveness of JSLS Metric. Validating the alignment between Δ_{JSLS} and physical reality is critical to justify DPA’s premise of using similarity as a proxy for probability. To this end, we first show that JSLS captures intrinsic distributional fingerprints.

Table 4: JSLS quantifies similarity between $D_{\text{base}} \sim \pi_{\text{own}}$ and datasets from π_{own} and π_{ind} , which can be distinguished via a threshold $\omega = 0.6$.

Dist.	Dataset	Semantic (m_r)	Lexical ($e^{k\gamma}$)	JSLS (Δ_{JSLS})
MedAlpaca (compared with $D_{\text{base}}^{\text{Wiki}}$, omitting superscript “wiki” later)				
π_{own}	D_{para}	0.984	1.121	0.811
π_{ind}	D^{Flash}	0.762	1.572	0.485
ChatDoctor (compared with $D_{\text{base}}^{\text{HCM}}$, omitting superscript “HCM” later)				
π_{own}	D_{split}	0.990	1.000	0.990
π_{own}	D_{para}	0.994	1.029	0.966
π_{ind}	D^{GPT}	0.727	1.238	0.587
CaseLaw (compared with $D_{\text{base}}^{\text{Del}}$, omitting superscript “Del” later)				
π_{own}	D_{split}	0.853	1.000	0.853
π_{own}	D_{para}	0.867	1.136	0.770
π_{ind}	D^{Ariz}	0.592	1.030	0.575

**Figure 7: The most relevant words across CaseLaw [23] Delaware and Arizona, measured by word relevance scores.**

As shown in Table 4, JSLS reliably separates proprietary provenance from independent sources across diverse domains. Empirical results demonstrate a robust separation margin: proprietary derivatives (including paraphrased variants) consistently yield high similarity scores ($\Delta_{\text{JSLS}} > 0.7$, highlighted in green), whereas independent datasets fall distinctly into the low-similarity zone ($\Delta_{\text{JSLS}} < 0.6$, highlighted in red). This discriminative power stems from JSLS’s dual modeling of the semantic-lexical correlated fingerprints.

The effectiveness of these intrinsic distributional fingerprints is founded on word relevance scores (Equation 4). Figure 7 shows the top relevant words in CaseLaw [23] data collected from Delaware and Arizona, respectively. Relevance scores highlight both shared domain terms (*plaintiff*, *defendant*) and unique distributional identifiers (*Delaware*, *Arizona*), with consistent scores between 0.2 and 0.4 reflecting their common legislative lexicon. In conclusion, word relevance scores reliably identify distributionally unique words, enabling robust distributional analysis via word preference.

Leveraging word relevance scores to weight semantic distributions, JSLS induces structural changes in the distribution geometry that filter out generic semantic noise, as visualized via PCA in Figure 8. This weighting reduces the superficial semantic similarity between ChatDoctor HCM proprietary data [19] and independent GPT synthetic data (e.g., from 0.811 to 0.729, a decrease of 10%), enabling the capture of discrepancies in semantic manifolds that unweighted baselines miss. Besides, JSLS quantifies linguistic style via the JS divergence of relevance distributions. Our analysis reveals that datasets from distinct sources may exhibit significant lexical divergence, as illustrated in Table 4 by the 1.572 gap between MedAlpaca’s wikidoc and flashcard [11] subsets, reflecting potential

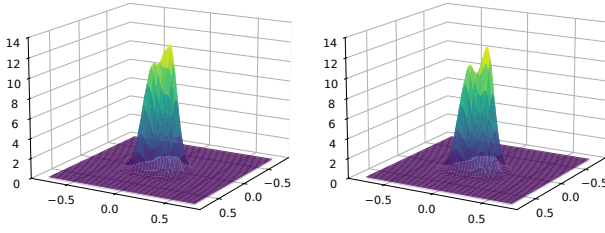


Figure 8: Semantic distributions of ChatDoctor [19] HCM conversations D_{base}^{HCM} via embedding K-means clustering, with principal components on the x- and y-axes and probability density on the z-axis. Left: Baseline distribution. Right: Relevance-weighted distribution.

fundamental terminological differences. Conversely, proprietary paraphrases consistently maintain moderate divergence. Therefore, by integrating these specific lexical signatures with weighted semantic analysis, JSLS reliably measures data distribution with intrinsic fingerprints, ensuring that high scores reflect genuine provenance rather than coincidental overlap.

7.2.2 Anchoring to Physical Utility and Privacy. To justify using JSLS as the ground-truth proxy for the DPA hypothesis test, we validate that it reflects practical fine-tuning by anchoring JSLS in two invariants enforced by the rational Trainer’s utility constraint: *semantic fidelity* and *privacy leakage*. As to semantic fidelity, Figure 9 benchmarks JSLS against MAUVE [33] (similarity via divergence frontiers) and MMD [20] (divergence via kernel projection). Correlations are measured by Pearson’s r , with significance p assessed via t-test. Our analysis reveals a robust statistical coupling: Δ_{JSLS} strongly aligns with MAUVE’s similarity measure ($r = 0.83$) while inversely mirroring MMD’s divergence metric ($r = -0.88$). Crucially, these correlations remain statistically significant ($p < 0.05$) despite limited samples, confirming that JSLS captures semantic consistency necessitated by model utility. Besides, the left of Figure 10 benchmarks JSLS against privacy overlaps via Personal Identifiable Information (PII) F1-scores [24]. We observe a distinct positive alignment where JSLS scores scale proportionally with privacy overlaps, effectively distinguishing low-risk independent data from high-risk proprietary baselines. Especially in the CaseLaw-paraphrase case, high PII retention ($F1 \approx 0.84$) persists despite lexical obfuscation, and JSLS correctly reflects this with a high score of 0.770. By grounding JSLS in these independent physical realities, we thereby legitimize its role as a valid proxy for distributional ground truth. Please refer to Appendix A.2 for detailed statistics.

JSLS’s alignment with reality highlights the critical limitations of existing baselines. Surface-level metrics (e.g., BLEU [31]) are vulnerable to false negatives via rank reversal, erroneously scoring obfuscated proprietary data lower than independent outliers, as shown in the right of Figure 10. The extended version [17] further shows via Bag-of-Words (BoW) [26] that the limitations of pure lexical metrics mirror the fundamental failure modes of standard MIAs. Conversely, pure semantic methods (e.g., MAUVE [33] and MMD [20]) suffer from false positives; as shown in Figure 9, they

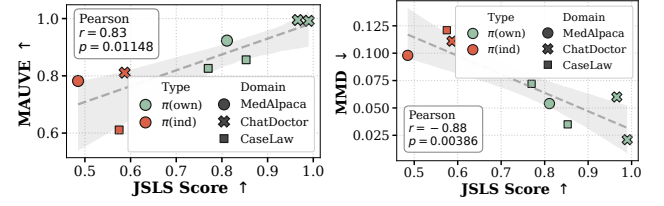


Figure 9: Anchoring JSLS to semantic reality, i.e. MAUVE [33] (left) and MMD [20] (right), respectively.

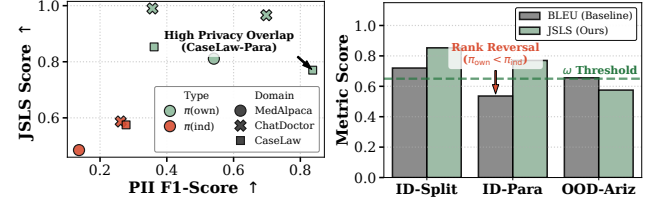


Figure 10: Comparisons between JSLS between privacy overlaps (left) and lexical baselines (right).

Table 5: Impact of fine-tuning tactics on the DPA solution: measured by mean and std of Extraction Deviation δ_{ext} .

Different Fine-tuning Tactics				
	No Tactic	Tactic 1	Tactic 2	Tactic 3
δ_{ext} mean	0.040	0.056	0.051	0.040
δ_{ext} std	0.043	0.067	0.060	0.039

Table 6: Impact of models and datasets on the DPA solution: measured by mean and std of Extraction Deviation δ_{ext} .

	Different Models			Different Datasets		
	Pythia	Qwen	Llama	MedAlpaca	ChatDoctor	CaseLaw
δ_{ext} mean	0.052	0.036	0.048	0.016	0.087	0.027
δ_{ext} std	0.060	0.035	0.052	0.016	0.059	0.020

assign proximally ambiguous scores to proprietary and independent domain data, blurring the provenance boundary. Furthermore, invasive interventions like dataset watermarking fail at data paraphrasing [35], leaving them ineffective against malicious trainers.

7.2.3 Utility-Informed Threshold Determination. To instantiate a suitable (Δ, ω) configuration for the *Dataset-Dataset Approximate Similarity* (Definition 4.1), we determine the decision threshold ω by anchoring it to the distinct separation observed between proprietary and independent distributions, as explicitly evidenced in Table 4. There, proprietary derivatives consistently score $\Delta_{JSLS} > 0.77$, whereas independent sources fall distinctly below 0.60. This divergence arises because utility necessitates fidelity: functional datasets consistently retain high JSLS scores regardless of obfuscation. Accordingly, we establish $\omega = 0.60$ as a conservative lower bound that maximizes robustness against evasion while strictly excluding false positives. This affirmatively answers *R1*: by validating ω as a *robust proxy* for the provenance ground-truth, we confirm that JSLS effectively captures intrinsic distributional fingerprints.

7.3 Audit Effectiveness and Robustness

7.3.1 Faithful Extraction via SIDE. With JSLs established, DPA’s validity hinges on whether the extraction method $\mathcal{G}$ faithfully recovers the latent model distribution π_M . We validate this extraction fidelity via *Extraction Deviation* (δ_{ext}). Ideally, provenance is verified by comparing the auditor’s reference D_{ref} directly to the root source D_{base} . However, since D_{base} is unobservable and potentially obfuscated, DPA relies on the extracted data $\mathcal{G}(M)$ as a proxy. SIDE’s objective is to ensure the extracted signal mirrors the ideal signal relative to D_{ref} . We quantify this alignment by defining δ_{ext} as:

$$\delta_{\text{ext}} = \|\Delta_{\text{JSLs}}(D_{\text{base}}, D_{\text{ref}}) - \Delta_{\text{JSLs}}(\mathcal{G}(M), D_{\text{ref}})\|_2 \quad (11)$$

A low δ_{ext} indicates that SIDE faithfully recovers the intrinsic fingerprints of the root D_{base} despite evasion, accurately projecting the latent dependency into the observable space for auditing.

Table 7 benchmarks this extraction fidelity. We observe that SIDE generally functions as a faithful extractor, maintaining small extraction deviation from the root source D_{base} across diverse evasion tactics. Notably, even under paraphrasing, SIDE faithfully recovers the latent semantic persistence (m_r) that survives moderate surface-level lexical penalty (e^{k_Y}) in the JSLs denominator, preserving the integrity of the provenance fingerprints. Crucially, this fidelity ensures reliable auditing: as evidenced by the bold values, the extracted provenance signals typically exceed the decision threshold ($\tau = 0.5$) while independent baselines fall below it. This confirms that SIDE effectively projects the latent discriminability of the unauthorized D_{base} into the observable space. We use 2K extracted and 2K reference samples by default; smaller sample budgets preserve the ordering between proprietary references and independent decoys, as shown in Appendix B.

We further quantify SIDE’s robustness via extraction deviation statistics. Table 5 confirms resilience against evasion: even under paraphrasing or distillation, deviations remain low on average ($\delta_{\text{ext}} \approx 0.040 \sim 0.051$), indicating reliable signal retention. Table 6 similarly demonstrates invariance to architectures and datasets, maintaining small deviations across diverse settings. Therefore, SIDE acts as a robust extractor that faithfully recovers intrinsic fingerprints, establishing a trustworthy foundation for the subsequent hypothesis test. Unless specified otherwise, subsequent experiments employ Qwen3-4B-Instruct [48] as the surrogate LLM.

7.3.2 Comparison with Baseline Extraction. To validate SIDE’s superiority, we compare it with five baseline extraction strategies using models fine-tuned on ChatDoctor ($D_{\text{trn}}^{\text{HCM}}$). Note that we include In-Distribution (ID) prompts as an *oracle* baseline to establish the theoretical performance upper bound, although they are basically infeasible in black-box audits. The baselines include:

- **Empty prompts:** Extraction via null inputs [49].
- **Tailored prompts:** Human-designed domain-specific queries (e.g., ‘Hi, doctor...’ in this case) [16].
- **ID prompts (Oracle):** Prompts directly sampled from the In-Distribution (ID) auxiliary training split $D_{\text{split}}^{\text{HCM}}$ [3].
- **OOD prompts:** Prompts from an Out-Of-Distribution (OOD) dataset within the same domain (e.g., D^{GPT}) [8].
- **Unrelated prompts:** Prompts from disjoint, unrelated domains (e.g. Alpaca-cleaned [41]) [3].

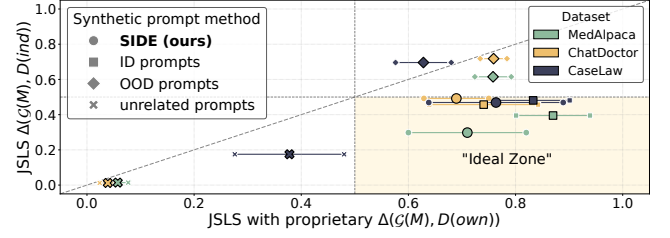


Figure 11: Visualizing prompt bias: Extracted distributions projected onto ID vs. OOD similarity axes. SIDE (circles) aligns with the oracle ID baseline (squares), whereas heuristic prompts introduce significant distributional shifts.

Table 8 reports the statistics of extraction deviation δ_{ext} quantified with JSLs. As expected, the oracle ID prompts achieve the lowest deviation ($\delta_{\text{ext}} \approx 0.046$) by directly querying with the target distribution. Crucially, SIDE ($\delta_{\text{ext}} \approx 0.054$) matches this oracle performance, significantly outperforming other heuristics. Conversely, unrelated prompts yield high deviation ($\delta_{\text{ext}} > 0.6$), as they force the model to generate generic content rather than the target medical distribution, confirming severe prompt-induced bias.

Figure 11 visually dissects this bias by plotting JSLs signals against π_{own} (x-axis) and π_{ind} (y-axis). We observe that oracle ID prompts (squares) define the ground-truth “ideal zone” in the bottom-right (high to π_{own} and low to π_{ind}). Conversely, baseline strategies exhibit distinct shifts: OOD prompts (diamonds) drift toward the upper-right due to distributional contamination, while unrelated prompts (crosses) collapse into the bottom-left, reflecting generic signal loss. Crucially, SIDE (circles) tightly overlaps with the oracle ID cluster. This alignment confirms that SIDE effectively mitigates prompt contamination, projecting the latent model distribution as faithfully as the original training data.

7.3.3 Hypothesis Testing Outcomes. Ultimately, DPA’s effectiveness depends on the consistency between the theoretical provenance label (b) and the empirical audit decision (b'), as formulated in Section 4. This requires a calibrated threshold pair (ω, τ) : ω defines distributional separation in the ground-truth space, while τ serves as the operational audit threshold on extracted samples. We measure this consistency using the DPA Advantage ($\text{Adv}_{\mathcal{A}}^{\text{DPA}}$, Eq. 1), derived from True Positive Rate (TPR) and False Positive Rate (FPR).

Tables 4 and 7 validate the calibrated setting ($\omega = 0.60, \tau = 0.50$). The threshold $\omega = 0.60$ separates proprietary sources from independent ones in the ground-truth distributional space, while $\tau = 0.50$ provides an operational boundary for black-box auditing with extracted samples. Table 7 thus reports point-estimate audit outcomes based on Δ_{JSLs} , where proprietary references are expected to exceed τ , whereas independent decoys should fall below it.

These decisions are further validated as non-parametric hypothesis tests using dataset-level bootstrap. For each replicate, we re-sample both extracted and reference corpora to recompute Δ_{JSLs} . DPA rejects H_0 only when the 95% lower confidence bound exceeds τ . Even for boundary cases in Table 7, the lowest-scoring positive case has a 95% confidence interval (CI) of $[0.5132, 0.6021]$, whose lower bound remains above $\tau = 0.5$; hence H_0 is rejected. In contrast, the hardest negative case yields a 95% CI of $[0.4615, 0.5261]$,

Table 7: DPA performance across diverse base models and evasion tactics. Report point-estimate provenance similarity Δ_{JSLs} and extraction deviation δ_{ext} (Eq. 11). Under $\tau = 0.5$, bold values mark point-estimate attribution ($\Delta_{\text{JSLs}} > \tau$), while *bold italics* mark point-estimate rejection of independent baselines ($\Delta_{\text{JSLs}} < \tau$). Bootstrap confidence validation is given in Section 7.3.3.

Models fine-tuned with No Tactic and Tactic 1: subset split.													
Dist.	Reference Dataset	Pythia-2.8B				Qwen3-4B-Instruct				Llama3-8B-Instruct			
		Sem. (m_r)	Lex. (e^{k_Y})	Δ_{JSLs}	δ_{ext}	Sem. (m_r)	Lex. (e^{k_Y})	Δ_{JSLs}	δ_{ext}	Sem. (m_r)	Lex. (e^{k_Y})	Δ_{JSLs}	δ_{ext}
MedAlpaca (compared with SIDE outputs from models fine-tuned on $D_{\text{trn}}^{\text{wiki}}$, omitting superscript “wiki” later)													
ID	$D_{\text{-}}\{\text{trn}\}$	0.785	1.003	0.783	< 0.001	0.874	1.138	0.768	0.002	0.837	1.049	0.798	< 0.001
ID	$D_{\text{-}}\{\text{para}\}$	0.747	1.106	0.676	0.018	0.834	1.389	0.600	0.045	0.793	1.283	0.618	0.037
OOD	$D^{\wedge}\{\text{flash}\}$	0.758	1.845	0.411	0.005	0.664	2.227	0.298	0.035	0.757	2.535	0.299	0.034
ChatDoctor (compared with SIDE outputs from models fine-tuned on $D_{\text{trn}}^{\text{HCM}}$, omitting superscript “HCM” later)													
ID	$D_{\text{-}}\{\text{trn}\}$	0.752	1.219	0.617	0.139	0.778	1.083	0.718	0.074	0.762	1.147	0.664	0.106
ID	$D_{\text{-}}\{\text{split}\}$	0.733	1.300	0.564	0.181	0.787	1.134	0.694	0.087	0.756	1.246	0.615	0.141
ID	$D_{\text{-}}\{\text{para}\}$	0.730	1.112	0.657	0.095	0.763	1.017	0.750	0.047	0.742	1.288	0.576	0.152
OOD	$D^{\wedge}\{\text{GPT}\}$	0.466	1.087	0.429	0.025	0.516	1.050	0.491	0.009	0.466	1.078	0.432	0.024
CaseLaw (compared with SIDE outputs from models fine-tuned on $D_{\text{trn}}^{\text{Del}}$, omitting superscript “Del” later)													
ID	$D_{\text{-}}\{\text{trn}\}$	0.759	1.073	0.708	0.021	0.892	1.004	0.889	0.001	0.763	1.240	0.616	0.056
ID	$D_{\text{-}}\{\text{split}\}$	0.736	1.102	0.668	0.034	0.826	1.012	0.816	0.001	0.773	1.272	0.608	0.060
ID	$D_{\text{-}}\{\text{para}\}$	0.721	1.035	0.696	0.005	0.661	1.238	0.689	0.007	0.804	1.002	0.802	0.001
OOD	$D^{\wedge}\{\text{Ariz}\}$	0.561	1.322	0.424	0.023	0.538	1.147	0.469	0.011	0.580	1.476	0.393	0.033
Models fine-tuned with Tactic 2: data paraphrase.													
Dist.	Reference Dataset	Pythia-2.8B				Qwen3-4B-Instruct				Llama3-8B-Instruct			
		Sem. (m_r)	Lex. (e^{k_Y})	Δ_{JSLs}	δ_{ext}	Sem. (m_r)	Lex. (e^{k_Y})	Δ_{JSLs}	δ_{ext}	Sem. (m_r)	Lex. (e^{k_Y})	Δ_{JSLs}	δ_{ext}
MedAlpaca (compared with SIDE outputs from models fine-tuned on $D_{\text{para}}^{\text{wiki}}$, omitting superscript “wiki” later)													
ID	$D_{\text{-}}\{\text{trn}\}$	0.739	1.166	0.634	0.031	0.862	1.051	0.820	< 0.001	0.856	1.147	0.746	0.004
ID	$D_{\text{-}}\{\text{para}\}$	0.646	1.035	0.625	0.034	0.840	1.030	0.816	< 0.001	0.854	1.014	0.842	0.001
OOD	$D^{\wedge}\{\text{flash}\}$	0.646	1.572	0.411	0.005	0.708	1.471	0.481	< 0.001	0.752	1.593	0.472	< 0.001
ChatDoctor (compared with SIDE outputs from models fine-tuned on $D_{\text{para}}^{\text{HCM}}$, omitting superscript “HCM” later)													
ID	$D_{\text{-}}\{\text{trn}\}$	0.698	1.254	0.556	0.188	0.718	1.077	0.667	0.104	0.726	1.292	0.562	0.183
ID	$D_{\text{-}}\{\text{split}\}$	0.706	1.344	0.525	0.216	0.725	1.082	0.670	0.102	0.732	1.298	0.564	0.181
ID	$D_{\text{-}}\{\text{para}\}$	0.736	1.129	0.652	0.098	0.755	1.010	0.748	0.047	0.758	1.16	0.655	0.096
OOD	$D^{\wedge}\{\text{GPT}\}$	0.425	1.011	0.421	0.028	0.469	1.050	0.447	0.019	0.454	1.014	0.448	0.019
CaseLaw (compared with SIDE outputs from models fine-tuned on $D_{\text{para}}^{\text{Del}}$, omitting superscript “Del” later)													
ID	$D_{\text{-}}\{\text{trn}\}$	0.685	1.008	0.680	0.030	0.736	1.148	0.641	0.045	0.813	1.189	0.684	0.029
ID	$D_{\text{-}}\{\text{split}\}$	0.679	1.005	0.676	0.031	0.703	1.103	0.638	0.046	0.797	1.206	0.661	0.037
ID	$D_{\text{-}}\{\text{para}\}$	0.742	1.272	0.583	0.035	0.719	1.013	0.710	0.004	0.813	1.189	0.684	0.007
OOD	$D^{\wedge}\{\text{Ariz}\}$	0.477	1.007	0.474	0.010	0.435	1.001	0.434	0.020	0.441	1.022	0.432	0.020
Models fine-tuned with Tactic 3: knowledge distillation.													
Dist.	Reference Dataset	Pythia-2.8B				Qwen3-4B-Instruct				Llama3-8B-Instruct			
		Sem. (m_r)	Lex. (e^{k_Y})	Δ_{JSLs}	δ_{ext}	Sem. (m_r)	Lex. (e^{k_Y})	Δ_{JSLs}	δ_{ext}	Sem. (m_r)	Lex. (e^{k_Y})	Δ_{JSLs}	δ_{ext}
MedAlpaca (compared with SIDE outputs from the model knowledge distilled via D^{flash} from a teacher fine-tuned on $D_{\text{trn}}^{\text{wiki}}$, omitting superscript “wiki” later)													
ID	$D_{\text{-}}\{\text{trn}\}$	0.763	1.021	0.747	0.004	0.786	1.043	0.754	0.003	0.842	1.046	0.808	< 0.001
ID	$D_{\text{-}}\{\text{para}\}$	0.701	1.059	0.662	0.022	0.781	1.199	0.651	0.026	0.786	1.248	0.629	0.033
OOD	$D^{\wedge}\{\text{flash}\}$	0.717	1.656	0.433	0.003	0.742	1.743	0.425	0.004	0.627	1.490	0.421	0.004
ChatDoctor (compared with SIDE outputs from the model knowledge distilled via D^{GPT} from a teacher fine-tuned on $D_{\text{trn}}^{\text{HCM}}$, omitting superscript “HCM” later)													
ID	$D_{\text{-}}\{\text{trn}\}$	0.754	1.189	0.634	0.127	0.785	1.144	0.686	0.031	0.783	1.147	0.682	0.008
ID	$D_{\text{-}}\{\text{split}\}$	0.719	1.166	0.616	0.140	0.761	1.210	0.629	0.130	0.779	1.154	0.662	0.108
ID	$D_{\text{-}}\{\text{para}\}$	0.687	1.002	0.686	0.078	0.776	1.047	0.741	0.051	0.774	1.147	0.741	0.051
OOD	$D^{\wedge}\{\text{GPT}\}$	0.497	1.090	0.456	0.017	0.468	1.084	0.432	0.024	0.492	1.119	0.439	0.022
CaseLaw (compared with SIDE outputs from the model knowledge distilled via D^{Ariz} from a teacher fine-tuned on $D_{\text{trn}}^{\text{Del}}$, omitting superscript “Del” later)													
ID	$D_{\text{-}}\{\text{trn}\}$	0.791	1.140	0.694	0.025	0.755	1.166	0.648	0.042	0.769	1.262	0.609	0.060
ID	$D_{\text{-}}\{\text{split}\}$	0.777	1.159	0.671	0.033	0.762	1.168	0.652	0.040	0.766	1.272	0.602	0.063
ID	$D_{\text{-}}\{\text{para}\}$	0.773	1.027	0.753	< 0.001	0.715	1.006	0.710	0.004	0.753	1.002	0.751	< 0.001
OOD	$D^{\wedge}\{\text{Ariz}\}$	0.618	1.359	0.455	0.014	0.588	1.287	0.457	0.014	0.552	1.318	0.419	0.024

Table 8: Impact of extraction algorithms on sampling fidelity: measured by mean and std of Extraction Deviation δ_{ext} . SIDE achieves fidelity comparable to the Oracle ID baseline.

Different Extraction Algorithms						
	Empty	Tailored	ID (Oracle)	OOD	Unrelated	SIDE(Ours)
δ_{ext} mean	0.226	0.193	0.046	0.074	0.683	0.054
δ_{ext} std	0.111	0.091	0.027	0.041	0.258	0.030

whose lower bound does not exceed τ , so DPA reports insufficient evidence rather than a positive attribution.

Figure 12 visualizes $\text{Adv}_{\mathcal{A}}^{\text{DPA}}$ over (ω, τ) , where positive detection requires both ground-truth separation ($\Delta_{\text{trn}} \geq \omega$) and operational evidence ($\Delta_{\text{ext}} \geq \tau$). The high-advantage “island” stably favors $\tau \leq \omega$ (especially $\tau \in [0.45, 0.60]$), consistent with information-bottleneck attenuation of extracted fingerprints. Outside this region, overly loose thresholds conflate independent distributions and reduce audit reliability. Therefore, this operational stability

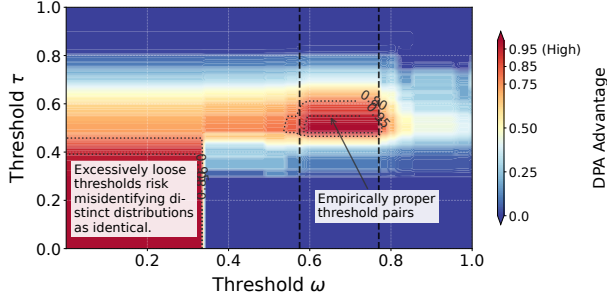


Figure 12: Analysis of DPA thresholds ω and τ for provenance.

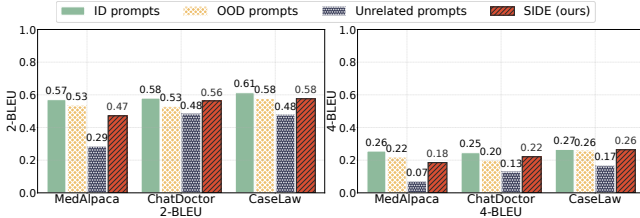


Figure 13: 2-BLEU (left figure) and 4-BLEU (right figure) for different data extraction methods.

affirmatively answers *R2*: by coupling SIDE’s latent distributional extraction with the robust JSLS metric, DPA achieves reliable audit performance across datasets and evasion tactics; averaged over the stable threshold region, it attains 95.62% TPR and 4.38% FPR.

7.3.4 Production-Model Evaluation. We further test DPA on a production fine-tuned gpt-4.1-nano [29] model trained with 3K MedAlpaca WikiDoc [11] samples for 5 epochs. Proprietary references, including paraphrased variants, obtain high Δ_{JSLS} scores (over 0.76), while OOD decoys remain separated (around 0.42). The ID proprietary audit yields a 95% CI of [0.82, 0.90], whose lower bound exceeds $\tau = 0.5$; the OOD decoy does not reject H_0 . This confirms DPA’s applicability to black-box production fine-tuned models.

7.4 Privacy Implications and Duality

Beyond data provenance, DPA reveals a fundamental duality: the mechanisms verifying data usage are structurally isomorphic to those performing privacy attacks. To answer *R3*, we now explore this “double-edged sword,” empirically demonstrating how high-fidelity fingerprints can be repurposed to facilitate privacy risks.

7.4.1 Data Extraction Attack. SIDE’s effectiveness in reconstructing training data is evident in Figure 13, where it achieves BLEU scores comparable to oracle ID prompts and outperforms unrelated prompts by over 30%, proving its ability to recover high-fidelity text without data access. This extends to sensitive information: Figure 14 confirms concrete privacy risks, as SIDE effectively extracts PII, achieving recall and F1-scores competitive with domain-specific baselines (e.g., > 20% recall), thus validating the structural tension between auditability and leakage.

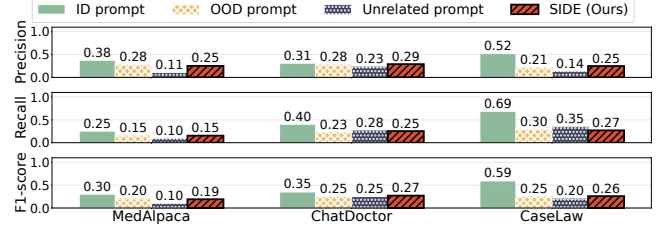


Figure 14: Performance comparison of Personal Identifiable Information (PII) extraction [24] across different data extraction methods, evaluating precision, recall, and F1-score.

7.4.2 Distribution Inference Attack. We demonstrate that the high-resolution alignment of DPA can be exploited for distribution inference attacks [14, 16]. By treating the JSLS score (Δ_{JSLS}) as a proxy for likelihood, an adversary can infer fine-grained properties of the training data, such as the exact mixing proportion of contributing parties, simply by maximizing the similarity score over a set of candidate reference distributions. Table 9 illustrates this vulnerability using models fine-tuned on varying mixtures of Delaware (D^{Del}) and Arizona (D^{Ariz}) CaseLaw data. The JSLS score exhibits a distinct diagonal dominance, consistently maximizing when the reference distribution’s composition aligns with the ground truth. For example, regarding a target model trained on a 25% D^{Del} mixture, DPA assigns the maximum score ($\Delta_{JSLS} = 0.823$) to the correct 25% reference, significantly suppressing scores for pure distributions (e.g., 0.647 for pure D^{Ariz}). This consistency allows adversaries to reverse-engineer the precise composition of private training sets, confirming that the high fidelity required for provenance auditing simultaneously exposes the model to granular property inference, both of which capture intrinsic distributional fingerprints. The extended version [17] further demonstrates that DPA resists differential privacy defenses, highlighting the associated privacy risks.

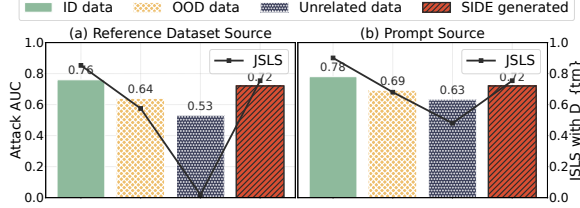
7.4.3 Auxiliary Dataset Synthesis and Selection. Sophisticated MIAs often rely on auxiliary datasets for membership calibration [3, 8]. DPA can exploit this dependency by enabling both the synthesis and selection of optimal reference data. Figure 15 reports the AUC of Reference-based MIA [3] on CaseLaw, comparing various reference datasets (left) and prompting strategies (right) against our SIDE baseline. All experiments utilize $D_{\text{base}}^{\text{Del}}$ as members and $D_{\text{split}}^{\text{Del}}$ as non-members. Specifically, SIDE-generated data (red, slashed) achieves an MIA AUC of 0.72, significantly outperforming OOD (≈ 0.64) and unrelated (≈ 0.53) baselines, effectively synthesizing high-utility attack materials that oracle ID data. Crucially, the overlaid plots reveal a strong positive correlation between the JSLS score (black line) and attack success (bars). This confirms that JSLS serves as a reliable predictor of attack effectiveness, allowing adversaries to strategically select, or synthetically generate, the most damaging auxiliary datasets solely based on distributional similarity scores.

8 Conclusion

This work presents *Distribution Provenance Audit (DPA)*, a robust post-hoc framework that anchors the audit in the fundamental utility constraint of LLM fine-tuning: the necessity to preserve *intrinsic distributional fingerprints*. DPA formalizes the audit as a

Table 9: The JSLS results $\Delta_{JSLS}(\mathcal{G}(M), D_{ref})$ in DPA enables distribution membership inference attack [14].

Fine-tuning D_{trn} Mix	Reference D_{ref} : D^{Del} Proportion				
	0%	25%	50%	75%	100%
25% D^{Del} + 75% D^{Ariz}	0.647	0.823	0.782	0.738	0.659
50% D^{Del} + 50% D^{Ariz}	0.632	0.830	0.839	0.769	0.747
75% D^{Del} + 25% D^{Ariz}	0.671	0.781	0.815	0.833	0.828

**Figure 15: The performances of reference-based MIA (Ref) [3] on CaseLaw while utilizing different reference dataset sources and prompt methods, respectively.**

distinguishability game with two technical innovations: (1) *Joint Semantic-Lexical Similarity (JSLS)*, which identifies integrated semantic and lexical signatures as a stable similarity proxy for provenance probability; (2) *Self-Instruct Data Extraction (SIDE)*, which reconstructs the model’s latent distribution without prompt-induced biases. Extensive evaluations demonstrate DPA’s reliability across diverse audit settings. Meanwhile, our findings uncover a critical *transparency paradox*: the high-fidelity evidence required for reliable provenance auditing is structurally isomorphic to mechanisms enabling privacy attacks, revealing an inherent tension between verifying Data IP and preventing privacy leakage.

In conclusion, this work proposes a paradigm shift toward auditing intrinsic distributional fingerprints, while revealing the dialectical tension between AI transparency and privacy risks. As a post-hoc audit, DPA assumes measurable distributional differences between proprietary and independent datasets; when they are statistically indistinguishable, or when a small proprietary fraction is heavily mixed into background data, provenance evidence becomes weakened or non-identifiable. Future work will extend DPA to multi-modal settings and more complex distribution mixtures, especially those integrating heterogeneous domain knowledge, to establish a comprehensive methodology for data provenance in the LLM era.

To support transparency and reproducibility, we maintain the core artifacts for *Distribution Provenance Audit (DPA)* at <https://github.com/Huangzirui1206/DistributionProvenanceAudit>. Extended details and results are available in the extended version [17].

Ethical Considerations

DPA is dual-use: the same mechanisms that enable provenance auditing may also expose privacy risks, as discussed in Section 6. To mitigate harm, we restrict experiments to public datasets and synthetic corpora generated via public APIs; no private user data is collected or processed. PII extraction is evaluated only quantitatively through aggregate F1 scores, without publishing sensitive content. We believe the benefit of enabling Data Owners to audit

unauthorized model training justifies this controlled study, while highlighting the need to jointly address auditability and privacy.

Generative AI Usage

In accordance with the ACM Policy on Authorship, we disclose that *Gemini* [6] and *ChatGPT* [1] were used to improve clarity, coherence, and presentation. *DeepSeek* [10] was used to generate paraphrased datasets for evasion evaluation. All AI-assisted content was reviewed and verified by the authors, who retain full responsibility for the work.

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A Complementary Experiment Results

A.1 Detailed Experimental Setup

All models are fine-tuned for 5 epochs with AdamW [22] (3×10^{-4} learning rate, batch size 128) on one NVIDIA A100 GPU. SIDE uses sentence-transformer [36], vec2text [28], and Qwen3-4B-Instruct [48], with $c = 2$, $k = 6/\ln 2$, $(\beta, \epsilon) = (0.5, 0.01)$, and $|D_{st}| = 100$. Following MAUVE [33], main evaluations use 2K extracted and 2K reference samples. Decoding uses temperature 0.7, top- $k = 30$, and repetition penalty 1.2. SIDE costs 25,248 tokens per 2K extracted samples and extracts 80/27.6 samples per minute for Pythia-2.8B/Qwen3-4B. Appendix B reports ablations.

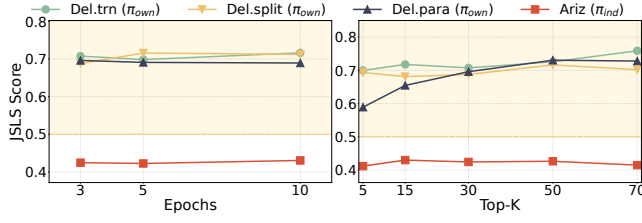


Figure 17: DPA results across fine-tuning epochs (left) and decoding top-k configurations (right), with light yellow regions indicating JSLS for models fine-tuned on π_{own} above the empirical threshold $\tau = 0.5$.

Table 10: Semantic, privacy, and lexical comparisons between π_{ind} and π_{ind} datasets across three domains.

Dist.	Dataset	MAUVE (Similarity $\uparrow$)	MMD (Distance $\downarrow$)	PII Extraction			BLEU-2
				Precision	Recall	F1-score	(Lexical $\uparrow$)
MedAlpaca (compared with $D_{\text{base}}^{\text{wiki}}$, omitting superscript "wiki" later)							
π_{own}	$D_{-} \{\text{para}\}$	0.923	0.054	0.784	0.413	0.541	0.454
π_{ind}	$D^{\wedge} \{\text{flash}\}$	0.782	0.098	0.571	0.078	0.137	0.435
ChatDoctor (compared with $D_{\text{base}}^{\text{HCM}}$, omitting superscript "HCM" later)							
π_{own}	$D_{-} \{\text{split}\}$	0.992	0.021	0.360	0.355	0.357	0.812
π_{own}	$D_{-} \{\text{para}\}$	0.995	0.060	0.640	0.768	0.698	0.646
π_{ind}	$D^{\wedge} \{\text{GPT}\}$	0.811	0.111	0.292	0.238	0.262	0.474
CaseLaw (compared with $D_{\text{base}}^{\text{Del}}$, omitting superscript "Del" later)							
π_{own}	$D_{-} \{\text{split}\}$	0.856	0.035	0.352	0.372	0.362	0.720
π_{own}	$D_{-} \{\text{para}\}$	0.826	0.072	0.919	0.769	0.837	0.536
π_{ind}	$D^{\wedge} \{\text{Ariz}\}$	0.611	0.121	0.266	0.291	0.278	0.656

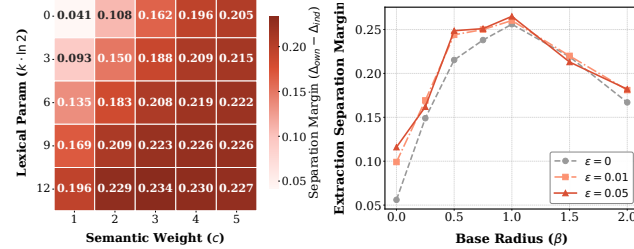


Figure 16: Ablation study of hyperparameters in JSLS (left) and SIDE (right).

A.2 Detailed Statistics for Metric Anchoring

Table 10 provides the granular statistics supporting the analysis in Figure 10 and Figure 9. It quantifies the divergence between JSLS and independent physical baselines: MAUVE (semantic similarity), MMD [20] (semantic divergence), PII Extraction (privacy), and BLEU (lexical). The data confirms that JSLS is the only metric that maintains consistent alignment with both semantic fidelity and privacy overlaps across both split and paraphrase.

B Ablation Studies

We further analyze the robustness of DPA to key hyperparameters:

Influence of JSLS hyper-parameters c and k . To verify the stability of DPA, we conducted a grid search over the semantic weighting factor $c \in [1, 5]$ and lexical bandwidth $k \in \{0, 3, 6, 9, 12\}/\ln 2$, adhering to

the JSLS validation in Section 7 (Figure 4). We measure performance using the *worst-case separation margin*, defined as $\min(\Delta_{\text{JSLS}, \pi_{\text{own}}} - \max(\Delta_{\text{JSLS}, \pi_{\text{ind}}}))$, the difference between the minimal JSLS score of proprietary datasets and that of similar but independent datasets. This strict metric quantifies the lower bound of audit robustness, ensuring that the least characteristic proprietary sample is still distinguishable from the most confusing independent outlier.

The heatmap in Figure 16 (left) provides a rigorous ablation path validating our joint design. First, the relevance-weighting strategy alone ($c = 2, k = 0$) significantly expands the separation margin from a negligible 0.015 (vanilla MAUVE in Table 10) to a distinct 0.108. Second, activating the lexical constraint ($k > 0$) further amplifies this margin monotonically, reaching 0.229 at $k = 12/\ln 2$, which confirms that the Laplacian kernel effectively acts as a “soft gate” to filter generic semantic noise. Regarding parameter selection, we empirically calibrate to ($c = 2, k = 6/\ln 2$). We choose this configuration because it maps JSLS scores to an interpretable dynamic range of $[0.4, 0.9]$. Crucially, this choice represents a stable operating point within the broad high-margin “plateau” visualized in the heatmap, confirming that DPA relies on robust design principles rather than fragile hyper-parameter tuning.

Influence of SIDE hyper-parameters β and ϵ . Ablation study of SIDE’s robustness is also evaluated using a Pythia-2.8B model fine-tuned on CaseLaw Delaware. We quantify extraction quality via the *Extraction Separation Margin*, defined as the minimal differential JSLS score $\min(\Delta_{\text{JSLS}}(D_{\text{ext}}, D_{\text{own}}) - \Delta_{\text{JSLS}}(D_{\text{ext}}, D_{\text{ind}}))$, where a higher value indicates fingerprints that are distinct from generic confounders. As shown in Figure 16 (right), the margin exhibits a distinct *inverted U-shape* across varying perturbation radii β , peaking around $\beta \approx 1.0$. Specifically, under-exploration ($\beta < 0.5$) yields a constrained margin ($0.1 \sim 0.15$), as restricted perturbations tend to generate generic legal phrasing lacking jurisdiction-specific signatures. Conversely, over-exploration ($\beta > 1.5$) degrades performance (< 0.18) by pushing embeddings off the valid semantic manifold into incoherent noise. The peak at $\beta \approx 1.0$ (margin > 0.26) represents the optimal region, where perturbations successfully uncover *intrinsic fingerprints*. Furthermore, the results demonstrate robustness to the stochastic range ϵ ; while performance is stable across values, a small non-zero noise ($\epsilon \in \{0.01, 0.05\}$) yields marginally superior separation compared to the baseline $\epsilon = 0$, confirming the benefit of bounded stochasticity in fingerprint recovery.

Influence of query/reference size. We evaluate JSLS on MedAlpaca with $N \in \{200, 1000, 2000, 5000\}$ extracted and reference samples. The relative ordering between ID and OOD remains stable across all sizes. Specifically, ID JSLS scores are $[0.939, 0.846, 0.855, 0.857]$, while OOD scores are $[0.408, 0.257, 0.246, 0.246]$. Scores stabilize beyond $N = 1000$, supporting $N = 2000$ as a robust default while indicating that fewer samples can often suffice.

Influence of fine-tuning epochs and decoding top-k. Experiment results on CaseLaw Delaware in Figure 17 reveal that the SIDE-based DPA solution performs consistently across different epochs and top-k choices, maintaining JSLS scores between D_{own} and SIDE-generated data D_{ext} around 0.7 for π_{own} and below 0.45 for π_{ind} . This confirms that DPA generates robust provenance results across varying fine-tuning epochs and decoding configurations.